Supplementary Material: Object Discovery from Spike Timing

Anonymous submission

1 Supplementary Tables

1.1 Released weights

These figures are given in prose in the main paper.

1.2 Training budget per baseline

1.3 Scaling with training data

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	CLEVR	Tetrominoes	Multi-dSprites
Slot Attention, matched budget	.2872 / .2220 / .0801	.1739 / .2192 / .1071	.1930 / .0876 / .0680
Slot Attention, released weights	.3747 / .2219 / .1165	.4742 / .2381 / .2355	.3205 / .0980 / .1204
Ours, matched budget	**.8421 / .7545 / .6410**	**.8580 / .7885 / .8173**	**.6974 / .8177 / .7628**

Table 1: Released weights ($\approx$32M exposures) against the matched-budget retraining ($100{,}000$ exposures) and against our model. Triples are ARI / F-IOU / O-IOU. All three rows are reported *without* the minimum-region-size rule, so the released and retrained Slot Attention rows are compared on identical terms; Table 1 of the main paper applies the rule to every model and our row there reads $.8408/.7528/.6387$, $.8568/.7950/.8170$ and $.6966/.8190/.7625$.

Model	Budget	Ours	Published
AKOrN (conv)	$1/210$	**60.5**	59.0 ± 4.3
CAE	$1/64$	**10.3**	4.0
AKOrN (attn)	$1/210$	60.4	91.0 ± 0.5
ItrSA	$1/210$	58.1	82.5 ± 1.7
CAE++	$1/64$	16.7	27.0
CtCAE	$1/64$	12.0	54.0

Table 2: Each baseline at our budget against its own published figure, by its own metric and evaluation code. *Budget* is the fraction of the published training schedule our matched-exposure protocol gives it. The top block is *not* budget-limited: AKOrN (conv) exceeds its published CLEVR number at $1/210$ of its schedule and CAE exceeds its own at $1/64$. The bottom block is.

Images	FG-ARI	FG-IOU	OBJ-IOU	Δ FG-ARI
10,000	.8421	.7545	.6410	—
11,000	.8426	.7539	.6427	$+.0005$
12,000	**.8449**	**.7607**	**.6461**	$+.0028$

Table 3: *Images* is the cumulative count of unique training images. Each row adds $1{,}000$ previously unseen CLEVR images for 10 epochs, resuming from the previous row. CLEVR evaluation.